321312

12425

3 Hours / 70 Marks Seat No. | | | | | | | | |

Instructions : (1) All Questions are *compulsory.*

(2) Illustrate your answers with neat sketches wherever necessary.

(3) Figures to the right indicate full marks.

(4) Assume suitable data, if necessary.

Marks

1. Solve any FIVE of the following : **10**

(a) Define fundamental and derived quantities.

(b) Justify the following equation whether it is dimensionally correct or not :

$$S = ut + \frac{1}{2}at^2$$

(c) The length of object is 18.2 ± 0.01 cm. Estimate percentage error in this measurement.

(d) State dimensions of (i) Area (ii) Work.

(e) State uses of (i) Yasti Yantra and (ii) Chakra Yantra.

(f) Define the term :

(i) Displacement

(ii) Velocity

(g) State any two kinematical equations of motion and state meaning of each term in it.

2. Solve any THREE of the following : 12

(a) Differentiate between longitudinal waves and transverse waves.

(b) Explain the term :

(i) Mass flow rate

(ii) Volume flow rate

(c) Define the following terms :

(i) Speed

(ii) Velocity

(iii) Average velocity

(iv) Acceleration

(d) Derive second kinematical equation of motion.

3. Solve any THREE of the following : 12

(a) Define work done and energy. Write its formula. State its M.K.S. units.

(b) Explain kinetic energy and potential energy with its examples.

(c) Define the term Power. Derive the relation between Power & work done.

(d) A student picks a book from the floor and put it onto a shelf 1.5 m high. The work done was 14.2 J. How much force was required to do this ?

4. Solve any THREE of the following : 12

(a) Calculate kinetic energy of 1500 kg car going at suburban speed of 14 m/s. and calculate potential energy of 2.0 kg hammer, when we moved it up 0.4 m ?

(b) State and explain “law of conservation of energy”.

(c) State and explain work-energy principle. State role of force in it.

(d) Define :

(i) Linear motion

(ii) Periodic motion

(iii) Uniform circular motion

(iv) Rotatory motion

(e) Differentiate between Streamline and Turbulent flow.

5. Solve any TWO of the following : 12

(a) Differentiate between centripetal force and centrifugal force with examples.

(b) Explain linear velocity and angular velocity with examples. Derive its relation.

(c) State principle of centrifuge machine. Explain its working and write its applications.

6. Solve any TWO of the following : 12

(a) Define surface tension. State relation between surface tension, capillary rise, radius of capillary tube. Calculate rise in level of water if capillary tube of diameter 1 mm in water and surface tension of water 0.07 N/m, density of water is 1000 kg/m^3.

(b) State Stokes law of viscosity. Derive equation for coefficient of viscosity by Stoke method.

(c) Define "Capillary Action". State its role for acute and obtuse angles and write its any two applications.

Define angle of contact and on what factors it depends.